

Section A: Basic Matplotlib Questions

Q1

Load the dataset and display the first 10 records.

Q2

Find:

- Number of rows
- Number of columns
- Column names
- Data types

Q3

Generate a line chart for the **Pulse** variable.

Q4

Generate a line chart for the **Maxpulse** variable.

Q5

Create a bar chart showing the average values of:

- Duration
- Pulse
- Maxpulse
- Calories

Q6

Generate a histogram for the **Duration** column.

Q7

Generate a histogram for the **Calories** column.

Q8

Create a scatter plot between:

- Duration and Calories
-

Section B: Intermediate Visualization Questions

Q9

Plot Pulse and Maxpulse on the same graph and add:

- Title
- Axis labels
- Legend
- Grid

Q10

Create a pie chart showing the top 5 most frequent workout durations.

Q11

Generate a box plot for the Calories column and identify potential outliers.

Q12

Generate box plots for all numerical variables.

Q13

Create subplots containing:

- Histogram of Pulse
- Histogram of Calories

within the same figure.

Q14

Create a scatter plot between:

- Pulse
- Calories

and interpret the relationship.

Section C: Data Analysis Questions

Q15

Determine:

- Minimum Calories burned

- Maximum Calories burned
- Average Calories burned

Visualize the findings appropriately.

Q16

Which workout duration occurs most frequently?

Support your answer using a graph.

Q17

Identify whether Duration and Calories have a positive relationship.

Generate an appropriate visualization and justify your answer.

Q18

Compare Pulse and Maxpulse distributions using:

- Histograms
 - Box plots
-

Q19

Find and visualize the average Calories burned for each Duration category.

Suggested chart:

Bar Chart

Q20

Which variable contains the highest variability?

Use:

- Statistical measures
- Box Plot

to justify your answer.

Section D: Outlier Detection Questions

Q21

Using Box Plot:

Identify which variable contains the most outliers.

Q22

Use the IQR method to detect outliers in:

- Duration
- Pulse
- Maxpulse
- Calories

Display the count of outliers for each.

Q23

Remove outliers from Calories and compare:

- Original Histogram
- Cleaned Histogram